

JKBOSE 2026

CLASS 11TH

Morphology of Flowering Plants

LECTURE-07

BOTANY

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Topics To Covered

1

Flower structure (contd..)

Part II

2

...

ADHESION OF STAMEN

Calyx $\rightarrow K$
Corolla $\rightarrow C$

- It is the attachment of Stamen with other floral parts.
- Two types:

① Epipetalous: Attachment of Stamen with Petal.

• ex. Brinjal

• represented by $\rightarrow \overset{\frown}{C}A$



② Epitopalous: Attachment of Stamen with tepals { whorl of tepals = perianth }

• ex. Lilly.

• represented by $\rightarrow \overset{\frown}{P}A$

COHESION OF STAMEN

• It is the attachment of Stamens among themselves.

• Two types:

① Polyandrous: All Stamens are free.
 • ex. Brassicaceae, Solanaceae.



② Adelphous: Filaments of Stamen are fused but anthers are free.



• 3 types:

→ Monoadelphous:

Filaments - fused } in single bunch
 Anthers - free



ex. China rose.

→ Diadelphous:

Filaments - fused } in two bunches
 Anthers - free



ex. pea plant
 $A (9) + 1$

→ Polyadelphous:

Filaments - fused } in many bunches
 Anthers - free



ex. Citrus

GYNOECIUM

"♀"

- It is ♀ reproductive whorl.
- Gynoecium consists of one/many Pistils or Carpels
- When Gynoecium contains Many Carpels, it is called as

Multicarpellary Condition.

02 types

Multicarpellary
Aporcarpous

↳ Many Carpels but free.
Ex. Lotus, Rose.

Multicarpellary Syncarpous

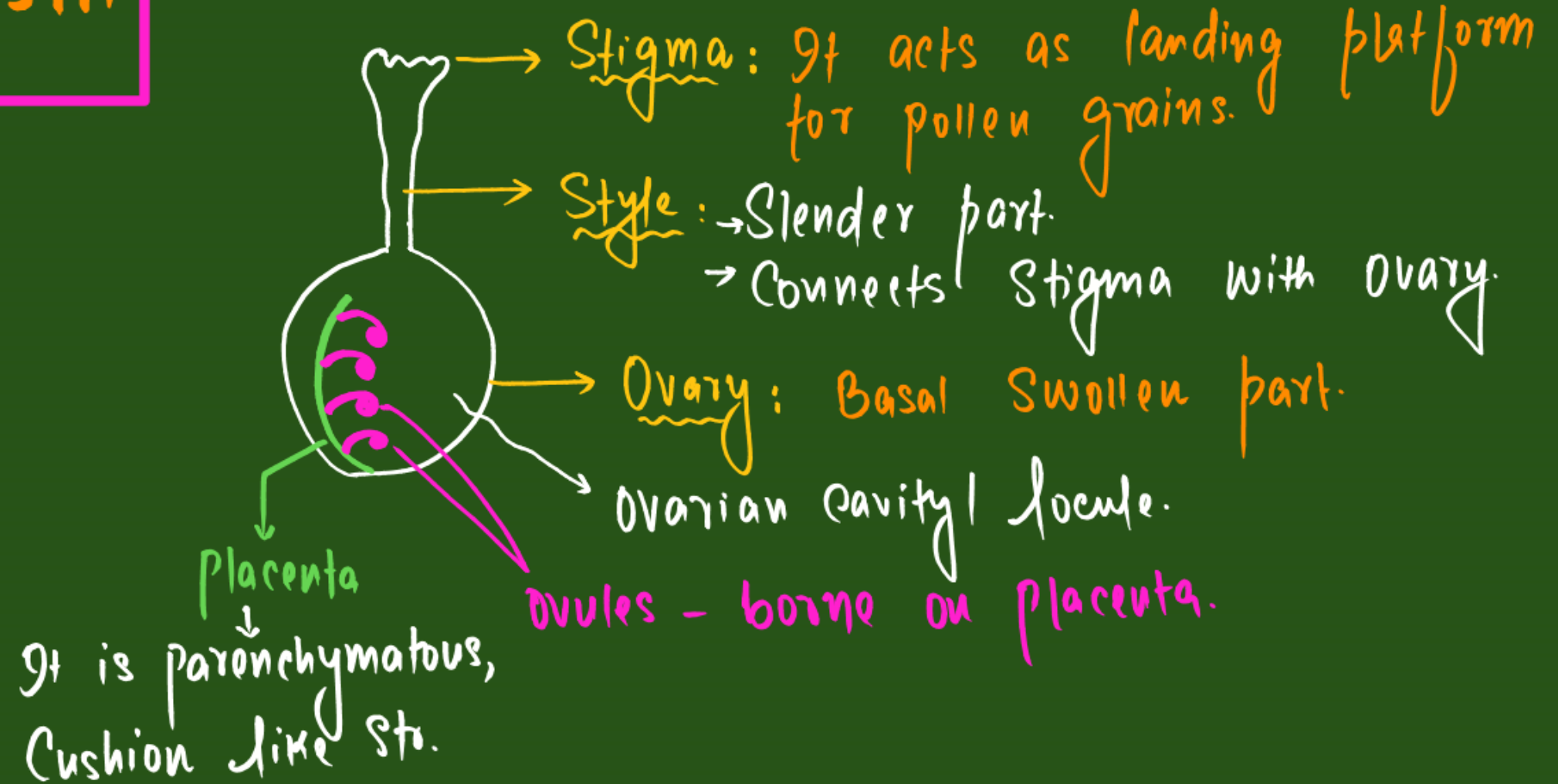
↳ Many Carpels but fused.

↳ Tomato, Mustard, Michelia, lady's finger.

Str. of CARPEL/pistil

→ 03 parts

- Stigma
- Style
- Ovary



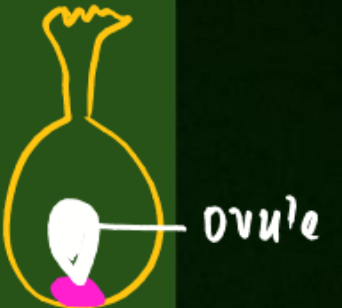
Note: After fertilisation, ovule forms seed and ovary forms fruit

PLACENTATION

→ It is the pattern of arrangement of Ovules on placenta.
→ 05 types:



MARGINAL	AXILE	PARIETAL	FREE CENTRAL	BASAL
Pea* → Ovules are borne on placenta along ventral Suture 	China Rose, Tomato, Lemon 	Mustard, Argemone 	Dianthus, Primrose, Chilly 	Sunflower, Marigold 



→ Here ovary is unilocular i.e., it possesses single locule.

→ Placenta is +nt axillary which bears ovules.
→ Ovary is Multilocular (Many locules) by formation of Septa.

→ Placenta is peripheral which bears ovules.
→ Initially ovary is unilocular but later becomes multilocular by False septum kld as Replum

→ Ovules are present on placenta is present at centre (freely)
→ Ovary is Unilocular.

Most advanced Placentation.
→ Single ovule is attached at base of ovary.

QUESTIONS

u/w

Match List-I with List-II.

(2024)

Choose the correct answer from the options given below:

- A** A-(IV), B-(III), C-(II), D-(I)
- B** A-(II), B-(III), C-(IV), D-(I)
- C** A-(II), B-(IV), C-(I), D-(III)
- D** A-(I), B-(II), C-(III), D-(IV)

Column-I	Column-II
(A) Rose	(I) Twisted aestivation
(B) Pea	(II) Perigynous flower
(C) Cotton	(III) Drupe
(D) Mango	(IV) Marginal placentation

QUESTIONS

u/w

Match List-I with List-II.

(2024)

Choose the correct answer from the options given below:

- A** A-(I), B-(II), C-(IV), B-(III)
- B** A-(III), B-(I), C-(IV), D-(II)
- C** A-(IV), B-(II), C-(I), D-(III)
- D** A-(IV), B-(I), C-(II), D-(III)

Column-I	Column-II
(A) Monadelphous	(I) Citrus
(B) Diadelphous	(II) Pea
(C) Polyadelphous	(III) Lily
(D) Epiphyllous	(IV) China-rose

QUESTIONS

u/w



Axile placentation is observed in

(2023)

- A** China rose, Petunia and Lemon
- B** Mustard, Cucumber and Primrose
- C** China rose, Beans and Lupin
- D** Tomato, Dianthus and Pea

QUESTIONS

μ/w



The term 'Polyadelphous' is related to:

(2016 - II)

- A** Corolla
- B** Calyx
- C** Gynoecium
- D** Androecium

HOMEWORK

A Questions



Thank
You